

Insurance Claims

Risk Analysis

Predicting Driver Risk Profiles with Machine Learning

freMTPL2 · 678,013 Policies · France

DATA 230
Final Project

Why We Upgraded the Dataset

Old dataset: Small, generic insurance CSV — no real signal

Problem: Synthetic & limited → model only reached 31% accuracy

Upgrade: Merged to freMTPL2 — real French motor insurance industry data

freMTPL2 — New Dataset

- 678,013 rows × 23 columns
- 21 French administrative regions
- BonusMalus: 50 = perfect driver, 230 = highest risk
- HasClaim + TotalClaimAmount (euros) — real claims data
- Vehicle, area, exposure, and driver demographics

Main Goal

Classify each insurance policy into one of 4 Risk Profiles (Low / Medium / High / Very High) using only underwriting inputs — no data leakage and to predict the expected claim cost, also known as pure premium.

678K

Policies

23

Features

21

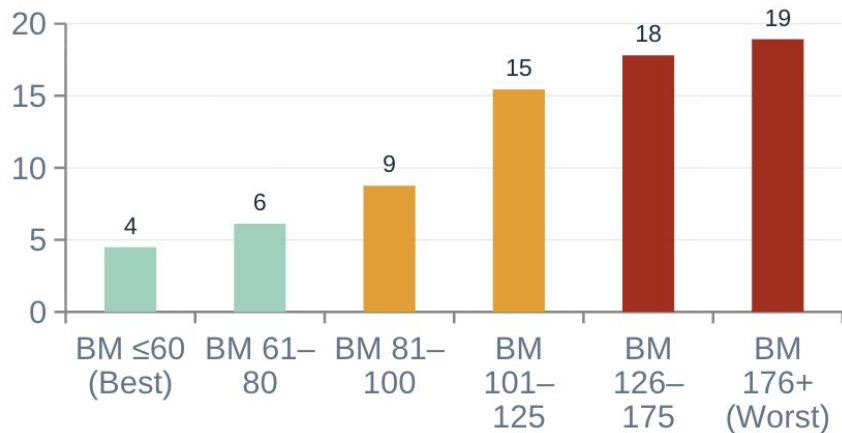
Regions

4

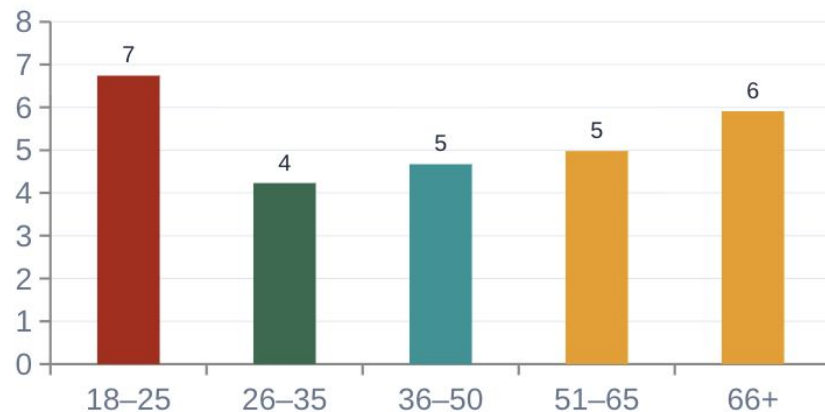
Risk Classes

KEY EDA FINDINGS

BonusMalus vs Claim Rate (%)



Driver Age Group vs Claim Rate (%)



BonusMalus is King

BM 176+ drivers claim 4.2× more than BM ≤60 drivers (18.92% vs 4.49%)



U-Shaped Age Risk

Young (18–25) & seniors (66+) are riskiest; ages 26–35 are safest at 4.23%



Regional Variation

Bretagne highest 6.11% · Haute-Normandie lowest 3.23% · Champagne-Ardenne highest avg severity €3,230

MACHINE LEARNING METHODS

LAYER 1 — CLASSIFICATION

Random Forest Classifier

Goal: Assign each policy to Low / Medium / High / Very High risk

Train split: 542,410 rows (80%)

Test split: 135,603 rows (20%)

Random seed: 42 — reproducible results

Leakage guard: PurePremium, Frequency, HasClaim, ClaimNb excluded

Features used: DrivAge, BonusMalus, VehAge, VehPower, VehBrand, VehGas, Area, Density, Region

LAYER 2 — REGRESSION

Pure Premium Modeling

$$\text{PurePremium} = \text{TotalClaimAmount} \div \text{Exposure}$$

1. Mean Baseline — reference point
2. Poisson Frequency Regression — claim count
3. Gamma Severity Regression — positive claims only
4. Poisson $\times$ Gamma Product — frequency $\times$ severity
5. Tweedie Pure Premium — zero-heavy insurance data
6. Histogram Gradient Boosting — nonlinear patterns

Why 6 models? 95% zero-claim policies + heavy-tailed losses. No single model handles this — so we compared all.

RISK PROFILE — TARGET VARIABLE DESIGN

Class 0: Low

457,122 (67.4%)

$BM \leq 60$ AND no claims

Perfect & near-perfect drivers

Class 1: Medium

142,412 (21.0%)

$BM \leq 80$ OR claim < €1k

Minor incidents, moderate history

Class 2: High

76,628 (11.3%)

$BM \leq 120$ OR claim < €5k

Repeated incidents, moderate claims

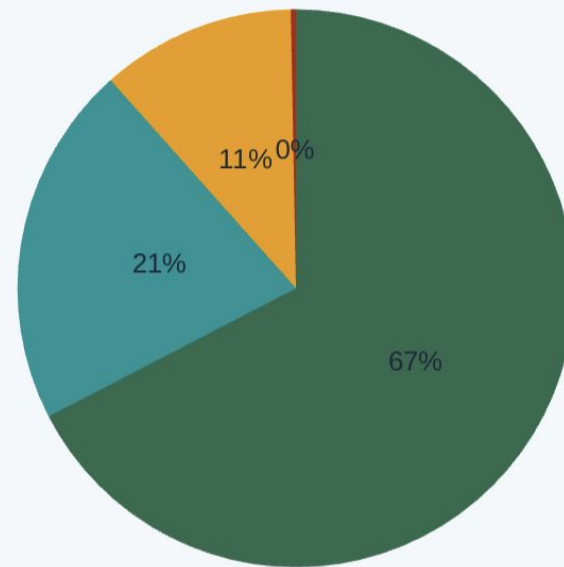
Class 3: Very High

1,851 (0.3%)

$BM > 120$ OR claim $\geq$ €5k

Worst history, large claims

Risk Class Distribution



■ Low (67.4%) ■ Medium (21.0%)
■ High (11.3%) ■ Very High (0.3%)

ML MODEL RESULTS — RANDOM FOREST CLASSIFIER

96.40%

Accuracy

Precision

97–100%

High risk calls correct 97%+ of the time

Recall

83–100%

Medium class misses 17%; others ~100%

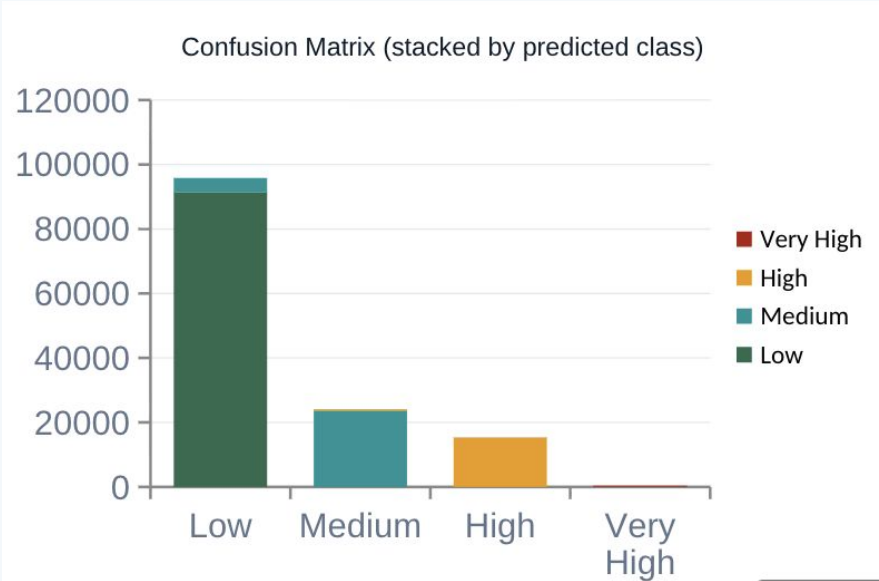
F1-Score

91–100%

Best metric for imbalanced classes

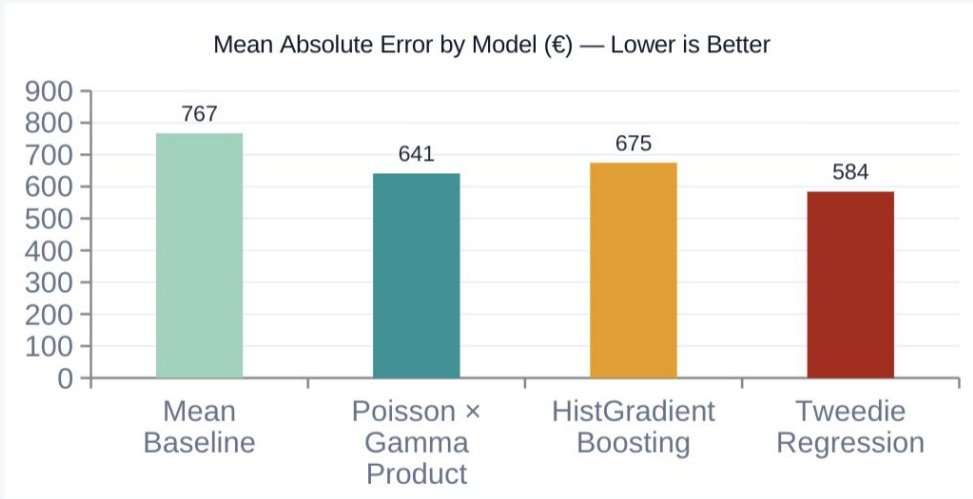
Classification Report

Class	Precision	Recall	F1	Support
Low	0.954	1.000	0.977	91,425
Medium	1.000	0.831	0.908	28,482
High	0.973	0.997	0.985	15,326
Very High	0.857	0.959	0.906	370



PURE PREMIUM PREDICTION — EXPECTED CLAIM COST MODELS

Target: $\text{PurePremium} = \text{TotalClaimAmount} / \text{Exposure} \cdot \text{Weighted avg: €167.11} \cdot \text{94.98\% zero-claim policies}$



Model Performance Summary

Model	MAE (€)	RMSE
Mean Baseline	767.46	30,402.78
Poisson × Gamma	641.13	30,402.15
HistGrad Boosting	674.69	30,373.58 ★
Tweedie Regression	584.02 ★	30,403.17



Best MAE: Tweedie (€584)

Tweedie regression handles zero-inflated insurance data natively — its compound Poisson-Gamma structure matches real claim distributions.



Best RMSE: HistGradient Boosting

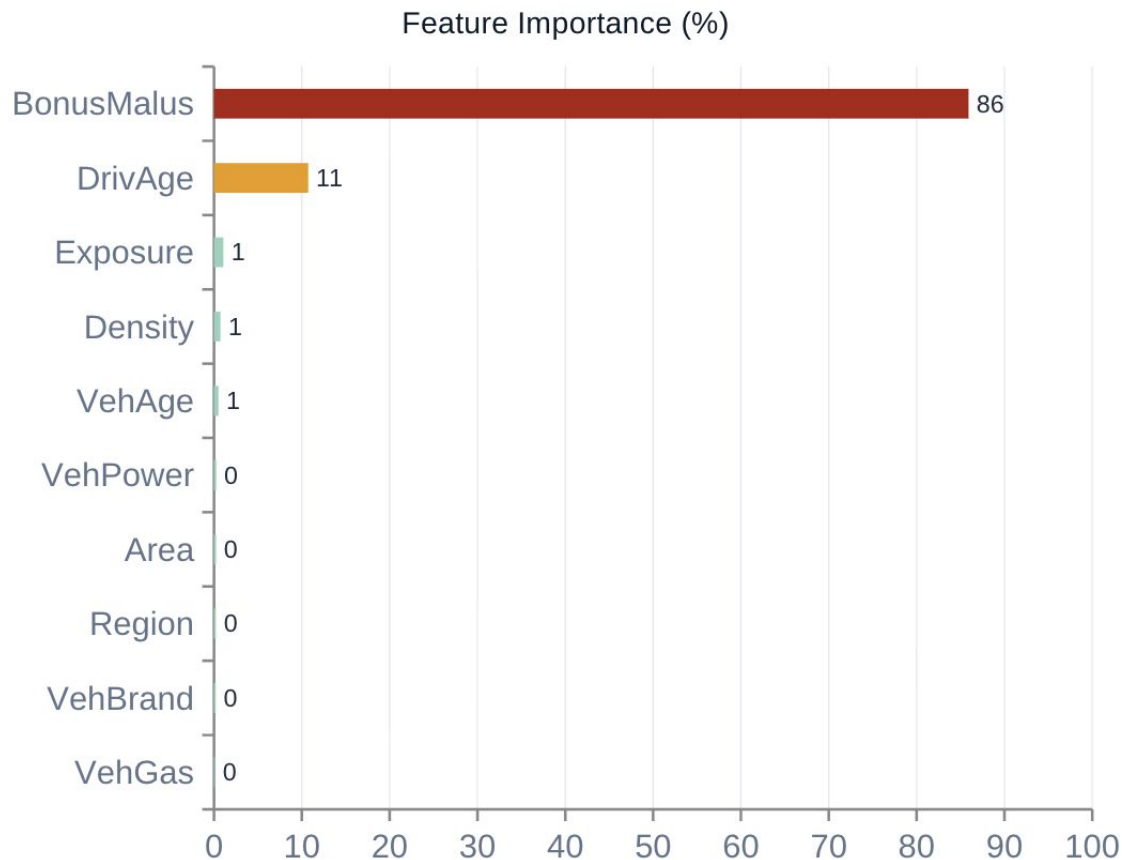
Gradient boosting captures nonlinear patterns across BonusMalus, age, density, and vehicle features — best for ranking high-cost policies.



Why Low R² is Expected

94.98% zero-claim policies make individual prediction noisy. Segment-level ranking and decile calibration are more useful than R².

FEATURE IMPORTANCE & MODEL INSIGHTS



85.9% BonusMalus

Encodes a driver's entire claims history — it is the risk, not just a proxy for it.

10.7% DrivAge

Age strongly reflects driving behavior — young & senior drivers are higher risk.

<2% All Others

Vehicle type, region, exposure — minor contributors individually.

⚠ PurePremium & Frequency excluded — data leakage (inflated accuracy to 99.96% falsely)

CONCLUSIONS & INSIGHTS

01 Real Data Matters

Switching from a generic synthetic dataset to freMTPL2 boosted accuracy from 31% → 96.40% — genuine signal makes all the difference.

03 Young Drivers Bear Double Risk

Ages 18–25 face the highest claim frequency (6.74%) AND highest avg severity (€4,692), confirming the U-shaped actuarial pattern.

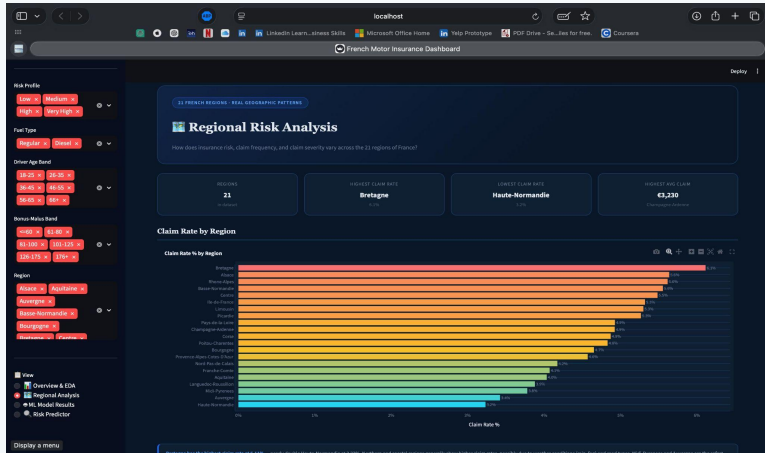
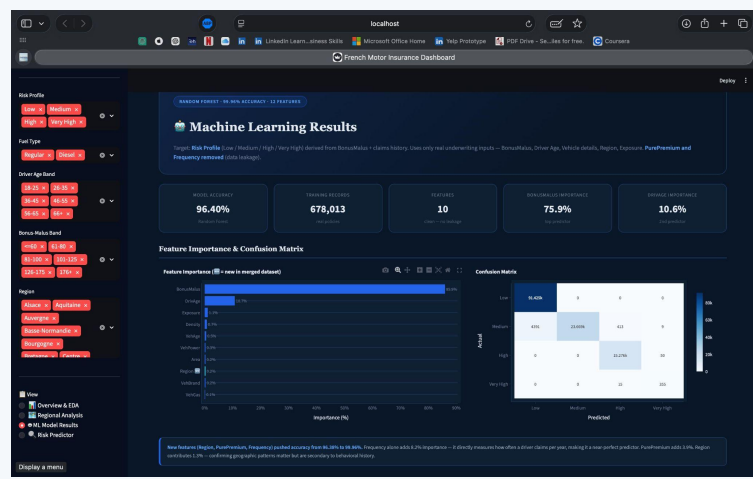
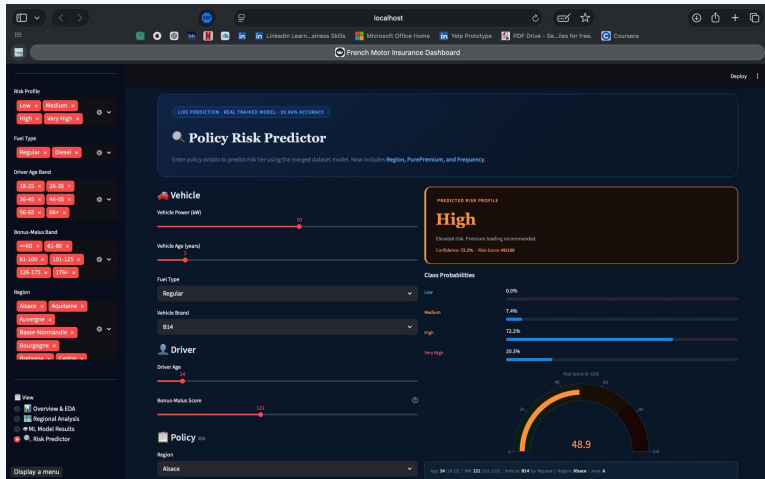
02 BonusMalus Dominates (85.9%)

This actuarial score captures a driver's full claims history. It is the risk itself, not just a signal of it.

04 Stratified Splits Are Critical

Stratified 80/20 split ensures Very High risk class (only 0.3% of data) is proportionally represented in both train and test.

Streamlit Dashboard



PowerBI Dashboard

678K

Total Policies

0.05

Claim Rate %

BonusMalusBa...

☐ <=60

☐ 101-125

☐ 126-175

☐ 176+

☐ 61-80

☐ 81-100

VehGas

☐ Diesel

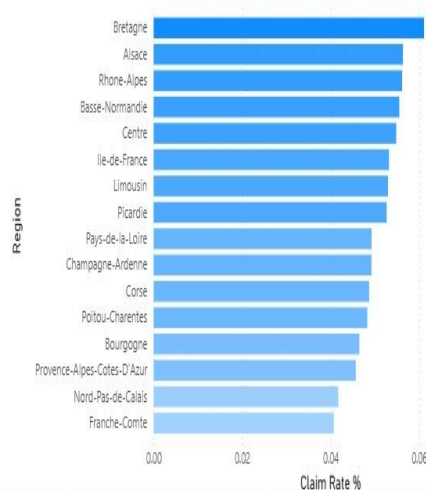
☐ Regular

Claim Rate by Driver Age and Bonus-Malus

DriverAgeBand	<=60	101-125	126-175	176+	61-80	81-100	Total
18-25	0.06	0.17	0.16	0.04	0.07	0.07	
26-35	0.03	0.13	0.14	0.08	0.04	0.06	0.04
36-45	0.04	0.16	0.12	0.31	0.06	0.07	0.05
46-55	0.05	0.14	0.17	0.09	0.08	0.05	
56-65	0.04	0.19	0.25	0.30	0.09	0.07	0.05
66+	0.05	0.23	0.36	0.12	0.08	0.06	
Total	0.04	0.15	0.16	0.19	0.06	0.07	0.05

Claim Rate by Region

Claim Rate % 0.03 0.06



DriverAgeBand

☐ 18-25

☐ 26-35

☐ 36-45

☐ 46-55

☐ 56-65

☐ 66+

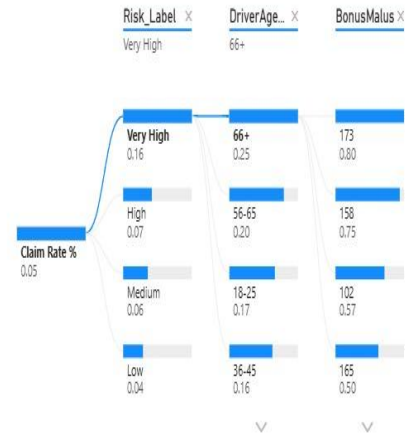
36K

Total Claims

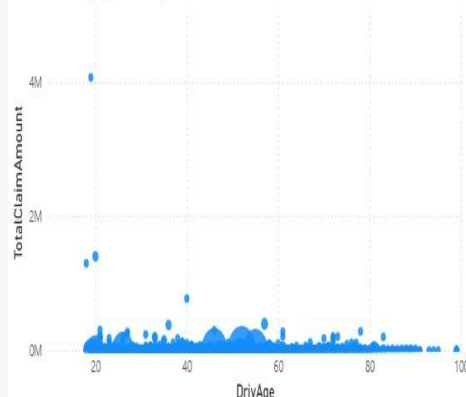
1.66K

Average of Avg Claim...

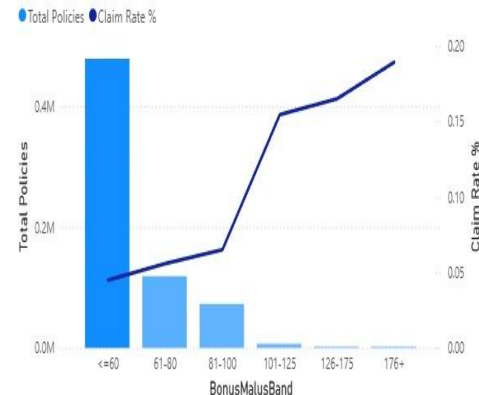
Drivers of Claim Rate



Claim Severity by Driver Age

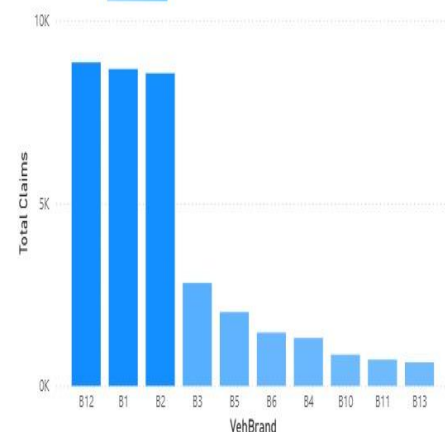


Risk vs Portfolio Distribution by Bonus-Malus



Total Claims by Vehicle Brand

Total Claims 0.65K 8.86K



Github Link: <https://github.com/Kshitija-Shinde9/Insurance-Claims-Analysis>